

Banking Mortgage Loan Approvals Project

Project Overview

- Analyst for Consumer Financial Protection Bureau (CFPB)
- Examine Approval Rates of Mortgage Loans
 - In NC, VA, MD, & DC
- Build predictive models for loan approval rates
- Use Power BI to help:
 - Determine Key Financial Metrics
 - Possible Demographic Drivers



Data Pipeline & Feature Engineering

Categorical Columns to One-Hot Encode:

- Activity_year
- Loan_purpose
- Debt_to_income_ratio
- State_code
- County_code
- Census_tract
- Occupancy_type
- Loan_type
- Derived_ethnicity
- Derived_race
- Derived_sex
- Applicant_age

Numerical Columns:

- Loan_amount
- Income
- Loan_to_value_ratio
- Interest_rate
- Loan_term
- Property_value

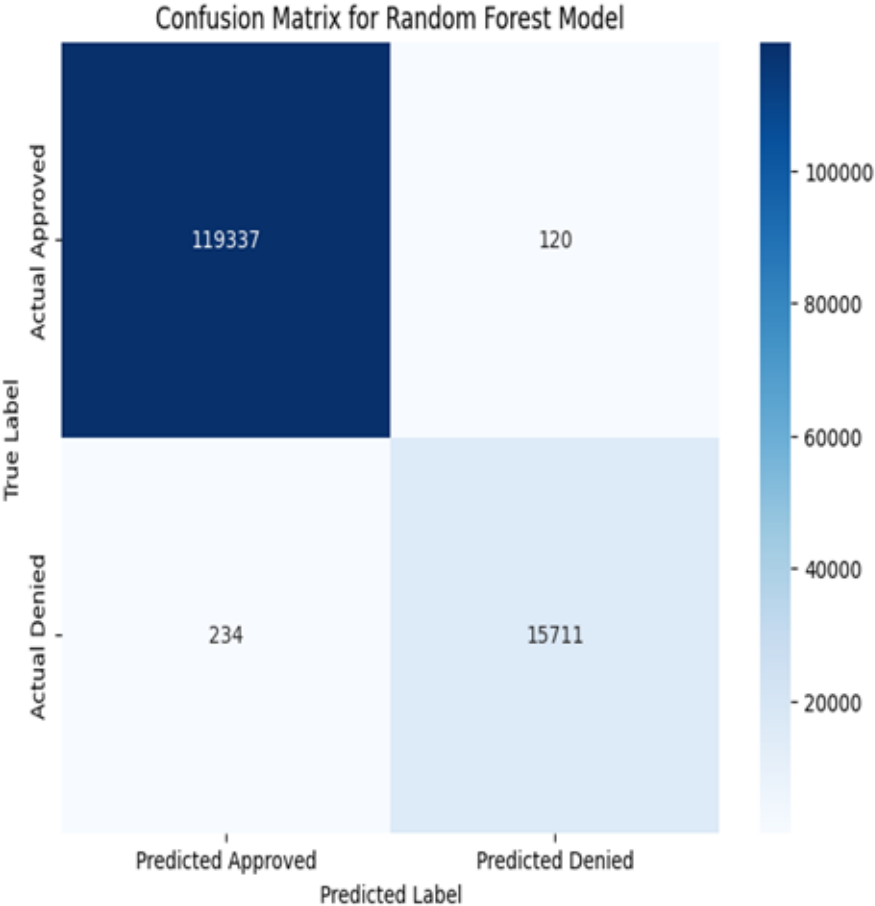
Replacement as NA:

- 8888
- Joint
- Demographics Not Available

Exploratory Data Analysis (EDA)

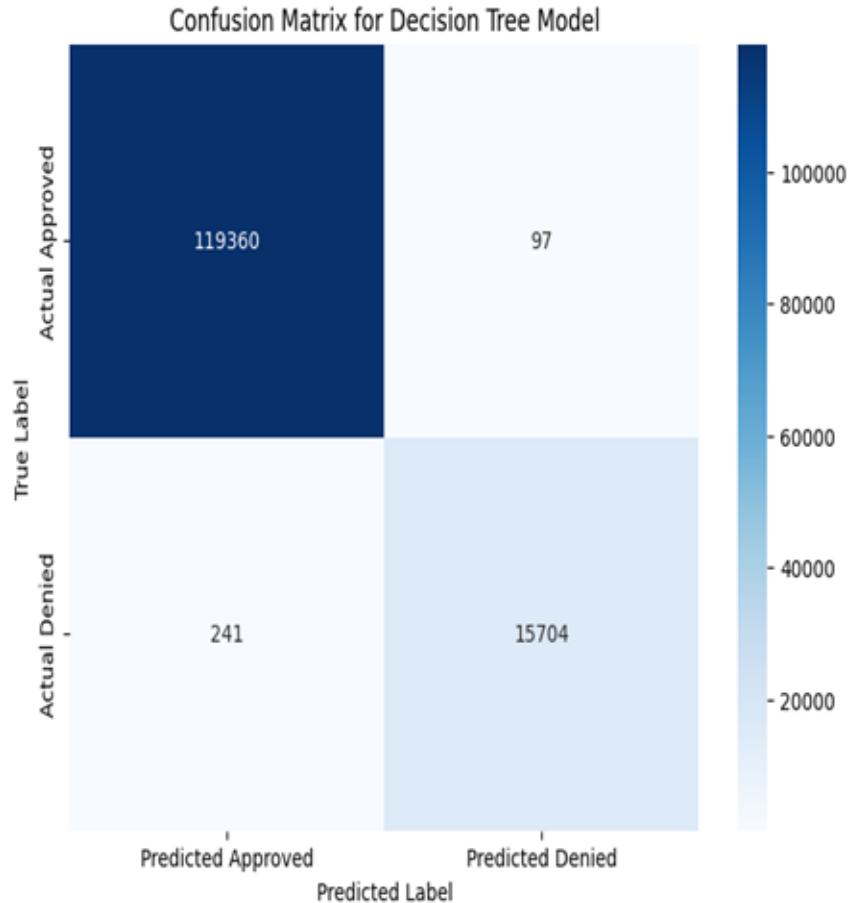
Rejected: 1
Accepted: 0

Confusion Matrix for Random Forest



- Accuracy: 99.74%
- Precision 99.24%
- **Recall: 98.53%**
- Predicted Applicants Approved: 25

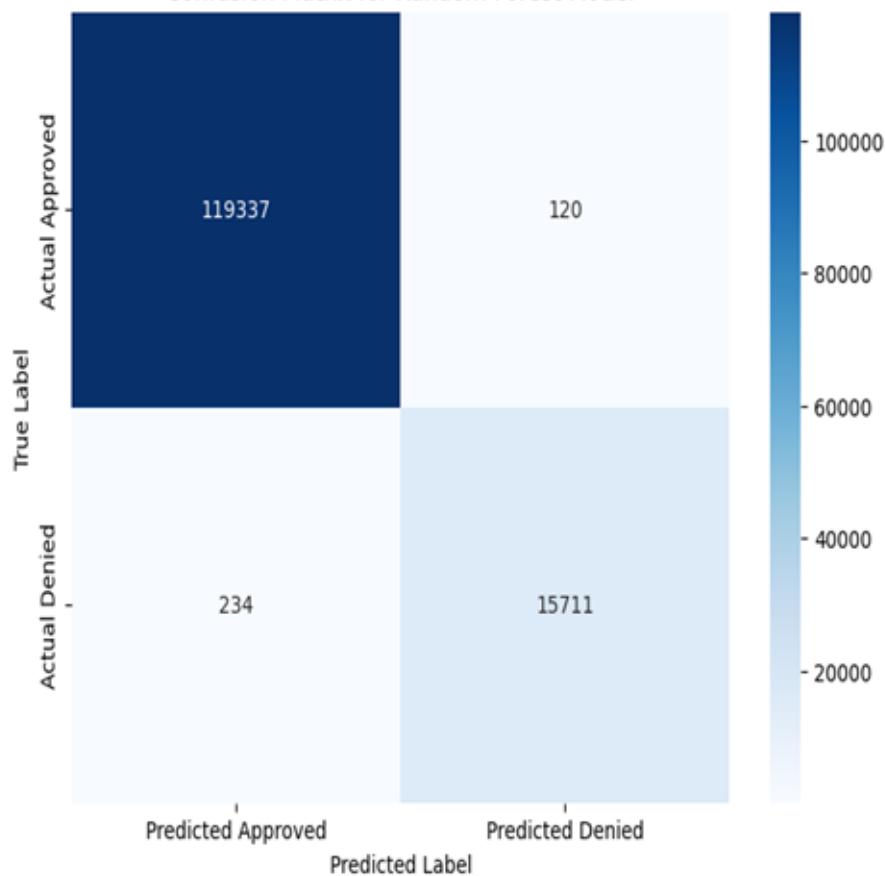
Confusion Matrix for Decision Tree Model



- **Accuracy: 99.75%**
- **Precision 99.39%**
- Recall: 98.49%
- Predicted Applicants Approved: 25
- A bit better to be **slightly** more precise

Which is Better?

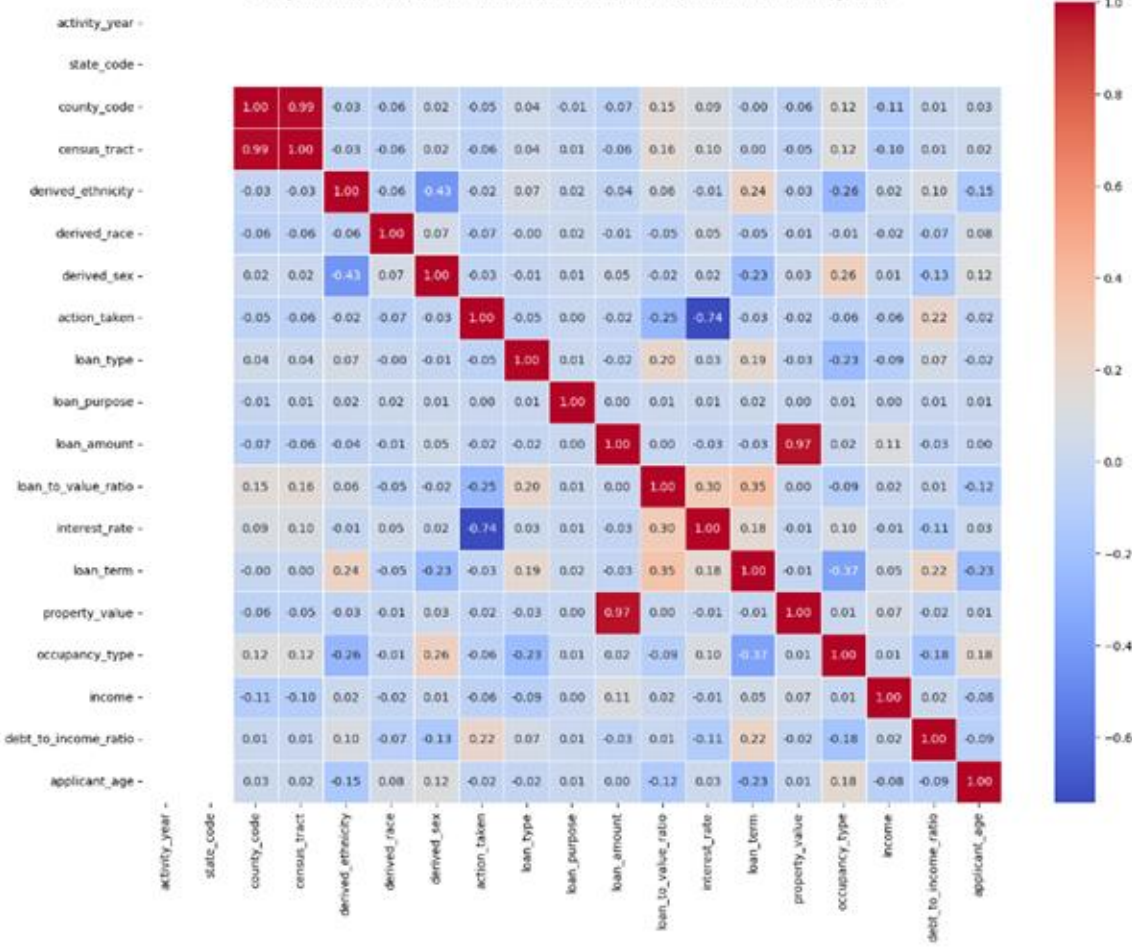
Confusion Matrix for Random Forest Model



- High Recall: Better for **not Overfitting**
- Better Generalization = Better at predicting future clients
- Both Relatively Close in Accuracy & Precision

Correlation Matrix

Comprehensive Pearson Correlation Matrix of All Features (Ordinal Encoded Categorical)



Positive Correlation:

- Interest rate & Loan to value ratio
- Loan term & Loan to value ratio

Strong Positive:

- Loan amount & property value

Negative Correlation:

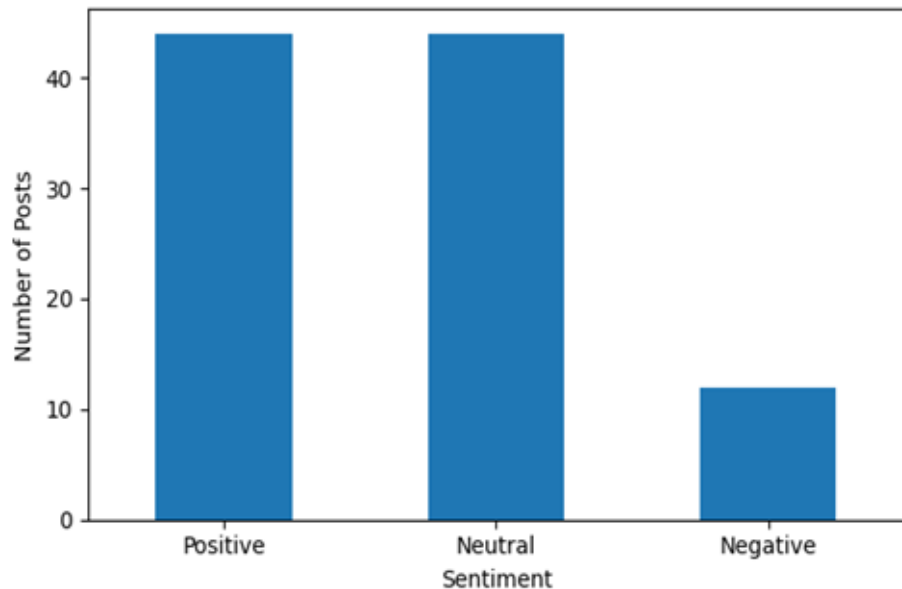
- Occupancy type & loan term

Strong Negative:

- Action taken & interest rate

Sentiment Analysis

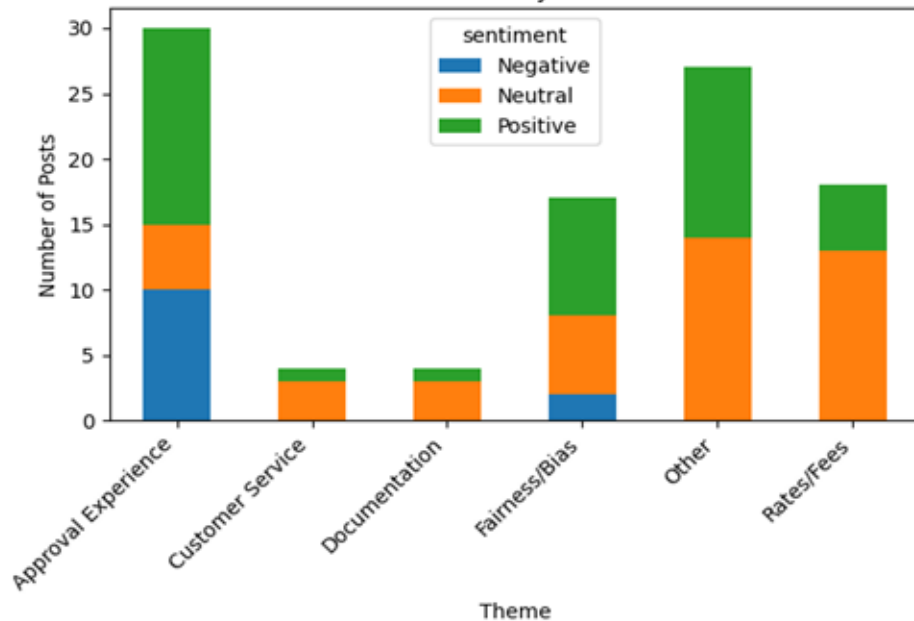
Overall Sentiment (n=100)



44% Positive - 44% Neutral - 12% Negative

The low negativity rate suggests the loan process is functioning well, but there are some issues from transparency and speed, not the approval decision itself

Sentiment by Theme



**Approval Experience shows strongest + sentiment
Rates/Fees and Fairness/Bias lean more neutral**

Positive sentiment is driven by approval speed, while areas like fees and fairness generate reveal where borrower feel the process could be clearer

Ethics of Predictive Learning

The models can unintentionally learn bias from the demographic data

- Model can see possible patterns from race, sex, or ethnicity and lead to discriminatory predictions

Sensitive personal data raises privacy concerns

- The HMDA includes income, demographics, and location, if this was mismanaged applicants could be profiled unfairly

Fair Lending laws restrict how demographic information can be used

- ECOA, Regulation B, and the Fair Housing Act require lenders to avoid discrimination which protects the consumers

Fairness checks and feature audits help prevent discrimination

- Reviewing model performance across groups ensures classes aren't harmed by unintended biases

Conclusion

- Best Model: Random Forest (High Recall; Less likely to Overfit)
- Predicted All Applicants to Approve
- Strong positive: Loan amount & property value
- Strong Negative: Action taken & interest rate

Power BI:

- Higher Debt/Collateral → Likely for Loan to Approve
- Lower Income → More Approval Chance → Higher Interest Rate for Loan
- Most applicants Avg income is very low

Sentimental Analysis:

- Low negativity → Mortgage process works
- Rates/Fees & Fairness/Bias → Neutral
- Positive Sentiment: Fast approvals